import streamlit as st

st.set_page_config(page_title="Ohm's Law Calculator", layout="centered")

st.title("⚡ Ohm’s Law Calculator")

st.write("Use this calculator to find Voltage (V), Current (I), or Resistance (R)")

User selection

option = st.selectbox(
"What do you want to calculate?",
("Voltage (V)", "Current (I)", "Resistance (R)")
)

Input fields based on selection

```
if option == "Voltage (V)":
current = st.number_input("Enter Current (I) in
Amperes", min_value=0.0)
resistance = st.number_input("Enter Resistance
(R) in Ohms", min_value=0.0)

if st.button("Calculate Voltage"):
voltage = current * resistance
st.success(f"Voltage (V) = {voltage:.2f} Volts")

elif option == "Current (I)":
voltage = st.number_input("Enter Voltage (V) in
Volts", min_value=0.0)
resistance = st.number_input("Enter Resistance
(R) in Ohms", min_value=0.1)

if st.button("Calculate Current"):
current = voltage / resistance
st.success(f"Current (I) = {current:.2f} Amperes")
```

```
elif option == "Resistance (R)":
voltage = st.number_input("Enter Voltage (V) in
Volts", min_value=0.0)
current = st.number_input("Enter Current (I) in
Amperes", min_value=0.1)

if st.button("Calculate Resistance"):
resistance = voltage / current
st.success(f"Resistance (R) = {resistance:.2f}
Ohms")

st.markdown("---")
st.caption("Formula: V = I × R")
```